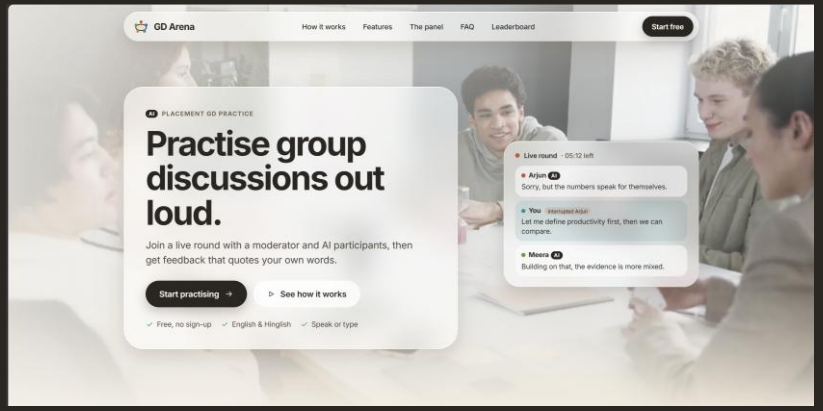
CODE2CAREER HACKATHON · AZISLY.AI × NIET

Problem Statement 2

GD Arena

Practise group discussions out loud with an AI panel that talks back, then get feedback that quotes your own words.



TEAM NEXORYX · Reg_48

Urvashi Team Leader · MCT, 3rd Year

Mehul Singh Member · CSE-IoT, 3rd Year

All other participants are AI

github.com/mehulsingh1/Codex

Most students never sit in a GD before the real one



6 to 8 people needed

Getting a full group together just to practise rarely happens.



A moderator, a topic

Someone has to run the round, keep time and close it.

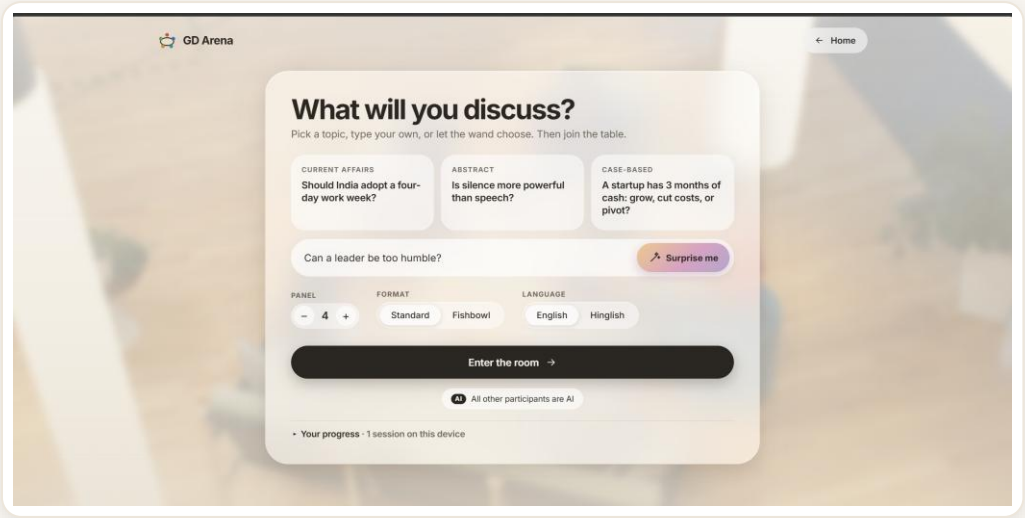


No honest feedback

Friends are polite. Nobody points to the moment you went wrong.

GD Arena gives every student that room, on demand: a moderator, 3 to 5 AI participants, a timer, and an honest report.

A full GD room, right in the browser



Pick a suggested topic, type your own, or hit Surprise me. Choose panel size, Standard or Fishbowl, English or Hinglish.

Voice-first
Speak, see live captions, hear the panel answer.

A panel with personalities
3 to 5 AI participants plus a moderator.

Real GD structure
Opening, timer, closing round.

Quote-linked report
Every point tied to your own words.

From topic to feedback in four steps

1

Set up

Topic list, custom topic or Surprise me. Panel of 3 to 5 AI, format, language.

2

Discuss

Moderator opens. You speak; the AIs answer you and each other, out loud.

3

Close

At the one-minute mark everyone gives a final statement.

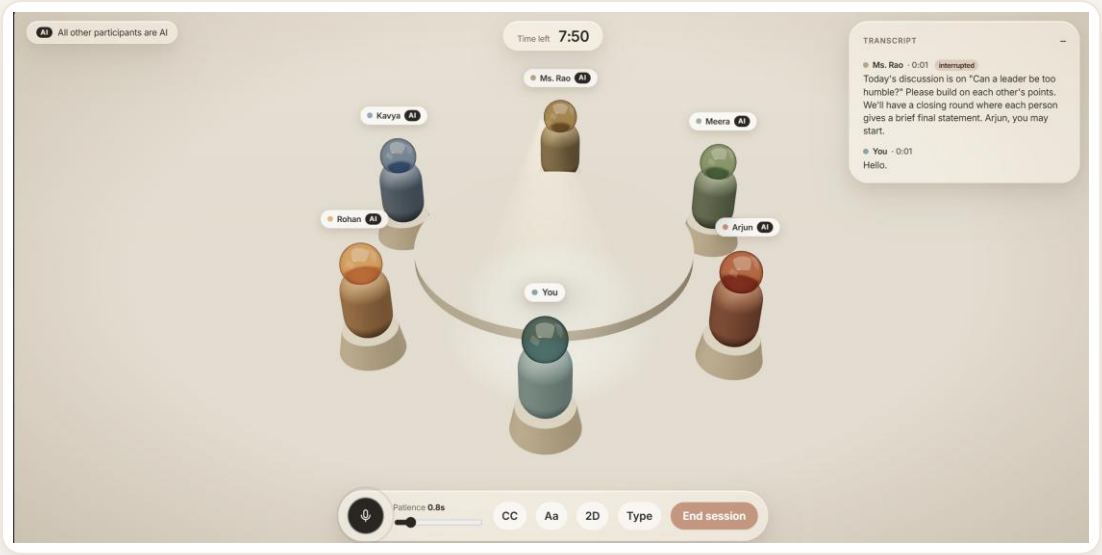
4

Report

Talk share, six scored areas, quotes, and what you could have said.

Then it sticks: each finished session updates your progress (score trend, strengths, habits) and earns leaderboard points.

The live room feels like a real table



Every seat carries an AI badge, and the room says it up front: all other participants are AI.

3D round table

A spotlight follows the speaker and pulses with their voice. 2D fallback.

Live transcript

Captions as you talk. Cut-off lines are tagged interrupted.

Your controls

Patience slider, caption size, Type fallback, End session.

Five personalities, each training a different skill

Ms. Rao
Moderator

Neutral and brief. Opens, keeps time, pulls quiet students in.

Arjun
The Dominator

Loud and certain, cuts in. Practise holding your ground.

Meera
The Data-driven

Backs everything with figures. Practise being specific.

Kavya
Quiet but sharp

Speaks rarely, lands one insight that reframes the room.

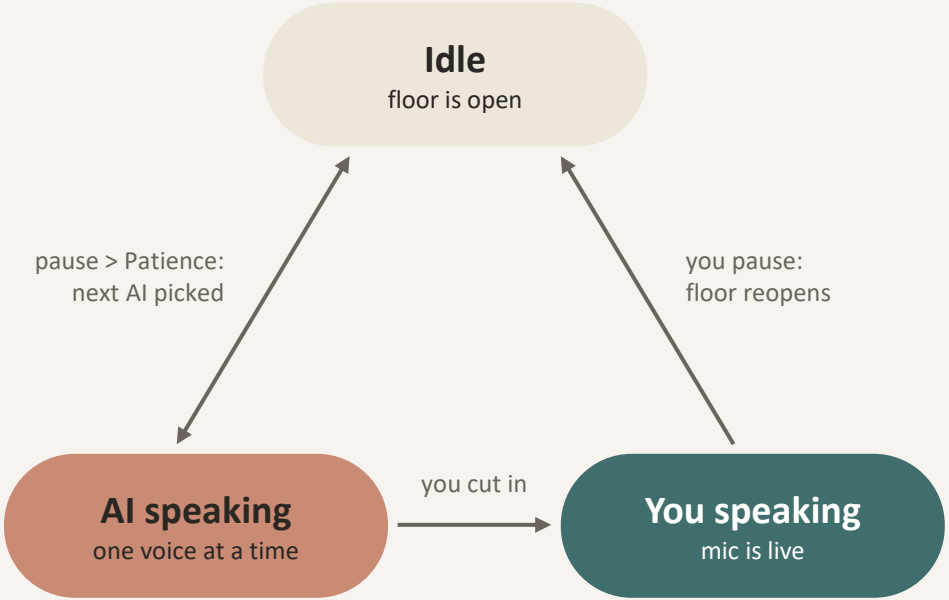
Rohan
The Drifter

Friendly tangents. Practise steering back to the point.

Sana
The Diplomat

Bridges views, invites others. Practise building on ideas.

Turn-taking: the server owns the floor



Barge-in stops audio instantly

The AI line is cut and tagged interrupted in the transcript.

AIs reply to each other

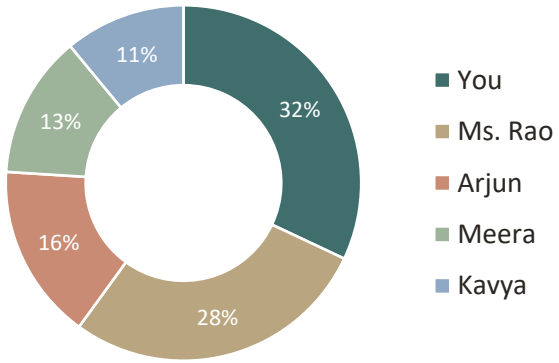
One LLM call per turn picks the next speaker and writes a 1 to 3 sentence line as strict JSON.

Adjustable patience

You set how long the panel waits after you pause (default 0.8 s).

A report that proves every point it makes

Who spoke how much



Sample session: four-day work week, panel of 3

Six scored areas

- Starting the discussion
- Quality of ideas
- Building on others
- Listening
- Handling interruptions
- Ending strongly

> *“Building on Meera, a four day week works best where output is measurable, so we should pilot it by sector.”*

Every feedback point links to a real line like this, with its timestamp. The code checks each quote against the transcript; invented quotes are repaired or dropped. Plus three “what you could have said” moments.

Architecture: one WebSocket, one LLM call per turn

Browser

React 19 · TypeScript · Vite

- Mic capture and voice activity
- Web Speech live captions
- Audio queue, barge-in cut-off
- Three.js table, 2D fallback
- Report and progress views

Backend

Python · FastAPI · WebSocket

- Session: floor, timer, transcript
- Turn engine: JSON validated by Pydantic
- Report builder and quote check
- Reconnect and resume
- 31 offline pytest tests

AI and voice

Groq · Whisper · Kokoro · Edge

- gpt-oss-120b for turns and reports
- gpt-oss-20b as fast fallback
- Whisper speech-to-text fallback
- Kokoro TTS on local GPU
- Edge TTS fallback and Hindi voices

Fast enough to feel live, sturdy enough not to break

Microphone denied
The Type bar appears automatically. The round carries on.

Connection lost
The client reconnects and the session resumes where it left off.

AI call fails
Retry once, then the fast model, then a moderator question.

Voice fails
Kokoro, then Edge TTS, then the browser's own speech.

Captions unavailable
Web Speech in Chrome, Groq Whisper everywhere else.

Low latency
One LLM call per turn. Speech plays sentence by sentence.

Extras that make practice stick

Fishbowl format

Watch the AIs debate first, then get invited in.

Hinglish room

Mixed Hindi-English panel with Hindi voices.

Accessibility

Live captions, text sizes, patience slider.

Missed openings

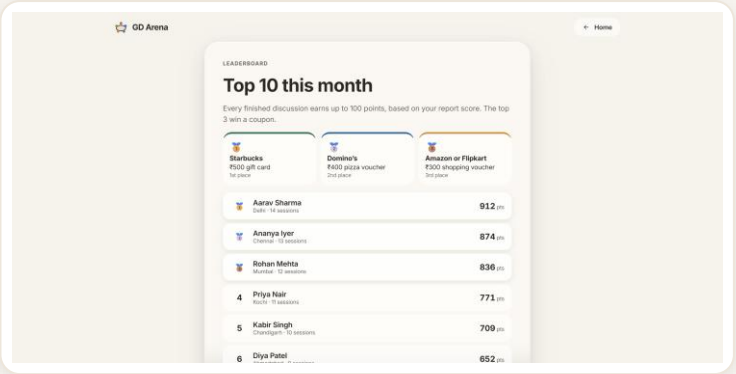
Replay of what you could have said.

Progress over time

Score trend, strengths, filler words, talk share.

Monthly leaderboard

Up to 100 points a session. Top 3 win coupons.



Starbucks ₹500 · Domino's ₹400 · Amazon or Flipkart ₹300. The demo board uses sample players; your own points stay in your browser.

Every core requirement, covered

Requirement	How GD Arena does it	
Set up a room	Topic list, custom topic, Surprise me; 3 to 5 AI plus moderator	✓
Live spoken discussion	Live captions; each AI answers out loud in its own voice	✓
Natural turn-taking	Waits for pauses, one speaker at a time, stops on barge-in, AIs reply to each other	✓
Structure	Moderator opens, countdown timer, closing round	✓
Report	Talk share and six scored areas, each linked to a verified quote	✓
Distinct participants	Separate instructions and separate voices per persona	✓
Speed and resilience	One call per turn; survives denied mic, dropped link, failed AI	✓
AI is disclosed	AI badge in setup, room, report and footer	✓

Thank you

Walk into your first real GD having already sat in a hundred.

github.com/mehulsingh1/Codex

Team Nexoryx · Urvashi & Mehul Singh · Reg_48

CODE2CAREER HACKATHON · AZISLY.AI × NIET

WHAT'S NEXT

- Friends in the same room, AI filling the empty seats
- A shared leaderboard with real accounts
- Fast live captions in every browser, not only Chrome